Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/07/2026
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kgibble@psu.edu	**Fax:**	FY 814 865 3604
PI Organization Type:	UNIVERSITY	**Phone:**	814 863 5343
Organization Name:	Pennsylvania State University		
PI Address 1:	104 Davey Laboratory 232		
PI Address 2:			
PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	09/30/2027
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2027 per the PI (Ed., 5/27/26). NOTE: New contract number is JPL 1724362 per the PI and J. Williams/JPL (Ed., 4/13/26). NOTE: New contract number is JPL 1631717 per J. Callas/JPL (Ed., 8/29/24) NOTE: End date changed to 09/30/2025 per J. Callas/JPL (Ed., 4/4/24) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1724362 ; JPL 1631717 ; JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The initial working uncertainty budget projected an accuracy of 2.2×10^(-16). The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of others, with a goal of reaching 1×10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy. Work will also include research on space optical clocks, including the PI participating as a member of NASA's science definition team (SDT) for optical clocks and time transfer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The group's work on microwave cavities produced an improved design for Cs and Rb fountain clocks. These are used in new clocks at the British, Canadian, South Korean, Polish, and Indian National Laboratories, and at CERN. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is now applied as a correction to International Atomic Time, including the correction for the PHARAO clock's frequency. More broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks, including testing fundamental physics with state-of-the-art optical clocks in space.
Task Progress:	ACES launched to the International Space Station (ISS) on April 21, 2025. In late 2020, a pre-launch accuracy evaluation of the PHARAO clock was published, describing the bulk of the Penn State pre-launch work for ACES. The important commissioning goals after launch are characterizing the mode of the state preparation cavity and optimizing the uncertainty evaluation of PHARAO’s collision and cavity phase frequency shifts. The ACES startup did not yield a signal from the Space Hydrogen Maser, requiring a significantly different approach for PHARAO’s collision and cavity phase frequency shift extrapolation for PHARAO’s on-orbit accuracy evaluation. Working with the ACES team, the Penn State group showed that PHARAO’s estimated frequency stability and the stability of the ACES microwave link can meet PHARAO’s accuracy goal with more averaging. These preliminary results will be updated through the commissioning period for ACES and its initial operation with additional NASA support. The Principal Investigator (PI) gave an invited talk at the NASA Biological and Physical Sciences (BPS) Fundamental Physics Program Analysis Group meeting and a talk on ACES was given at the 26th International Conference on Laser Spectroscopy. A number of talks on the first phases of the ACES mission are currently scheduled for 2026.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Abstracts for Journals and Proceedings	Cacciapuoti L, Pataraia S, Peignier T, Plumaris M, Weinberg S, Bischoff U, Crescence P, Helm A, Kehrer J, Lachaud R, Mitschke D, Niedermaier T, Esnault FX, Léger B, Thulliez E, Massonnet D, Goujon D, Pittet J, Perri A, Wang Q, Liu S, Schaefer W, Schwall T, Prochazka I, Schlicht A, Schreiber U, Laurent P, Lilley M, Roze J, Wolf P, Gibble K, Salomon C. "ACES in orbit." 26th International Conference on Laser Spectroscopy (ICOLS 2025), Elba Island, Italy, June 2-7, 2025. Abstracts. 26th International Conference on Laser Spectroscopy (ICOLS 2025), Elba Island, Italy, June 2-7, 2025. https://www.icols2025.it/wp-content/uploads/2025/05/BookofAbstracts_30may2025.pdf , Jun-2025